Human Body Communication: A Living Survey of Integrated Circuits and Channel Models

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v3.11 | 13 November 2026 | Literature snapshot: 13 November 2026

Abstract

This manuscript provides a fixed, searchable edition of the SparcLab human body communication (HBC) literature catalogue. Its purpose is to collect the publications in one place and make progress in integrated-circuit (IC) energy efficiency visible through publication-year plots. The 13 November 2026 snapshot contains 337 records: 103 IC publications, 147 channel/model publications, and 87 other references. All records are listed in a compact bibliography with stable identifiers and source links where available. IC plots begin in 2006, the earliest year represented in the IC collection, and display transmitter, receiver, and combined transmitter–receiver energy separately. Machine-readable bibliographic and measurement files accompany the manuscript. The resource supports publication discovery and inspection of archived IC performance data.

1 The catalogue at a glance

The catalogue brings together the records maintained by the SparcLab HBC survey website. Readers can search the PDF by title, first author, venue, year, or record identifier, and follow a linked title or DOI to its source. The accompanying data retain full stored author strings, additional source links, and measurement notes. Table 1 reports the total collection sizes.

Table 1: Total records in the 13 November 2026 catalogue.

Collection	Records	Location in this manuscript
Integrated circuits	103	Section 5
Channels and models	147	Section 6
Other literature	87	Section 7
Total	**337**	Complete catalogue

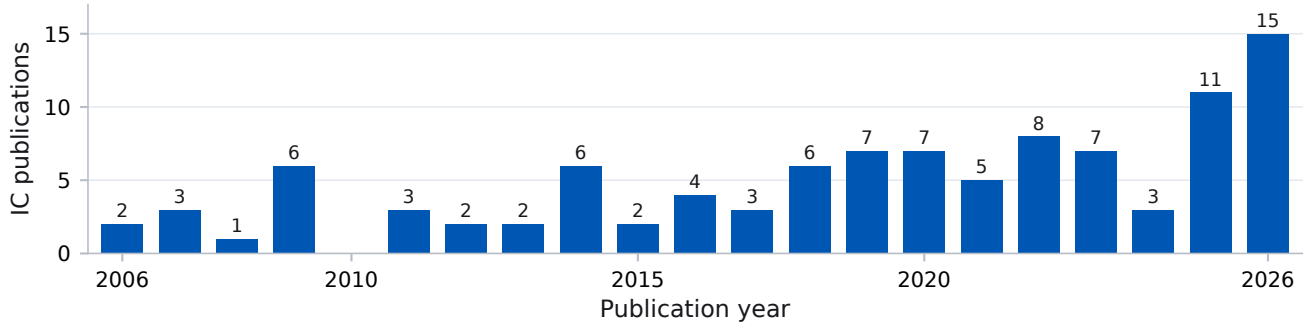


Figure 1: IC publication counts by stored publication year, starting with the earliest included IC papers in 2006. Counts total 103 publications; conference and journal reports are counted as separate records when both are retained. The 2026 count covers the snapshot through 13 November 2026.

The collection includes HBC and related body-area/implant literature. Its three categories support finding papers; only records classified as eligible fabricated custom solid-state ICs enter the comparison plots. These are counts of publications, not counts of unique chips. The catalogue is a curated collection and does not claim exhaustive coverage of the field.

2 IC energy efficiency over time

Figure 2 provides a visual record of the energy values in the IC collection. Each point is a stored operating point; one publication can contribute more than one point. Lower energy per bit indicates greater energy efficiency within the stated measurement scope.

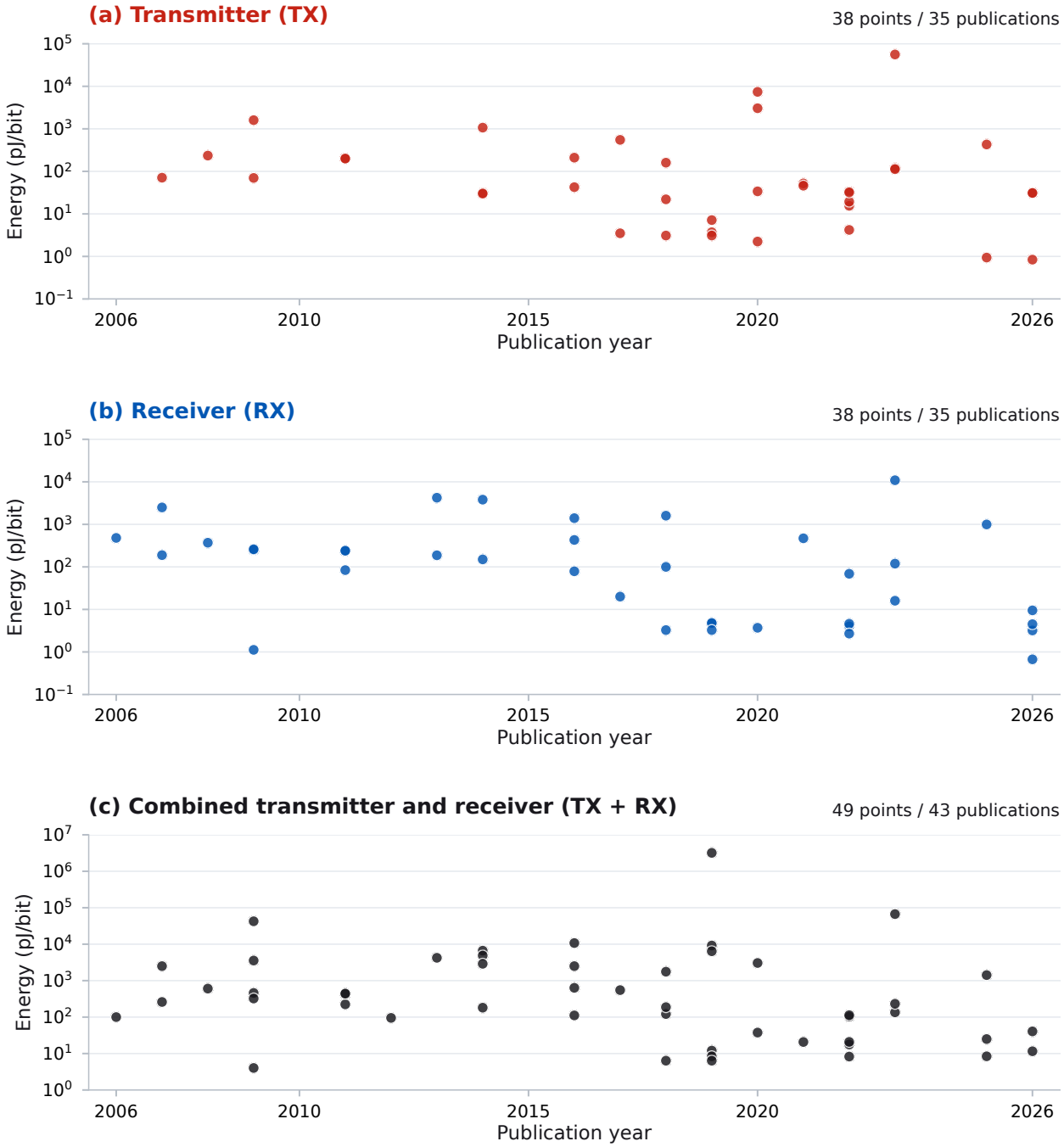


Figure 2: Archived IC energy per bit versus publication year: (a) TX, (b) RX, and (c) TX+RX. The plots use positive numeric entries from the same eligible IC records as the website. Vertical axes are logarithmic and have panel-specific ranges. Combined values are those reported or explicitly calculated in the archive. Missing values are omitted; separate panels can therefore contain different publications. Overlapping points can conceal one another. Source identifiers and values for every plotted point are supplied in `anc/energy_points.csv`.

3 Using and citing this edition

Reading the plots. The plots retain stored energy scopes, operating conditions, and publication years. The IC archive includes adjacent body-area and implant modalities as well as electrical HBC. Data rates, component exclusions, and link configurations differ between papers; consult the original publications and record notes before making a quantitative comparison. No trend fit or performance ranking is imposed. Four inherited combined-energy entries (158, 184, 186, 303) are withheld because the stored total is below a listed component value. Original values are retained in the data pending source reconciliation.

Finding a paper and its data. Sections 5–7 contain all 337 records, grouped by category and ordered from recent to earlier years. Red reference numbers connect entries to the full dataset. Titles are retained in full; long author lists are abbreviated. There are 335 DOI links with the full DOI strings printed. The source package includes `catalogue.bib`, `anc/catalogue.csv`, the full `anc/survey_snapshot.json`, `anc/energy_points.csv`, and scripts to regenerate the catalogue and figures. Missing metadata and measurements remain explicit.

Citation. Cite the original paper for its technical result. Suggested resource citation: Shreyas Sen, "Human Body Communication: A Living Survey of Integrated Circuits and Channel Models," v3.11, 13 November 2026; literature snapshot: 13 November 2026. The companion online resource is the SparcLab HBC survey website.

4 Release history

Version / date	Release and conference context
V1 19 June 2022	Original launch during Shreyas Sen's invited talk, "Secure and Efficient Internet of Bodies (IoB) using Body as a 'Wire'." in the "Human Body Communications" workshop at the IEEE Radio Frequency Integrated Circuits Symposium (RFIC).
V2 23 February 2023	Updated survey released during Shreyas Sen's invited talk, "Secure and Efficient Internet of Bodies (IoB) using Body as a 'Wire'." in Forum 7, "Advanced Circuits and Technologies for Wearable and Implantable Devices," at the IEEE International Solid-State Circuits Conference (ISSCC).
v3.10 October 2026	Expanded catalogue, compact bibliography, scope screening, and disclosure updates.
v3.11 13 November 2026	Current website and catalogue manuscript release; publisher-verified additions and preserved disclosures. It includes the literature search and verified backfills completed on 13 November 2026.

This revision preserves the prior removal of 12 out-of-scope or author-excluded records; stable reference numbers are retained. Version naming now uses `v<major>.<month>`; `.11` denotes November. The full release date records the year.

AI usage and conflict of interest

AI usage statement. Artificial intelligence (AI), including large language models accessed through OpenAI ChatGPT and Codex, has been used for literature discovery, bibliographic organization, and preparation of the website, manuscript, and scripts. AI-assisted content may contain mistakes. If you find a mistake, please let us know.

Conflict-of-interest statement. Shreyas Sen is a founder, Chairman and Chief Technology Officer of Quasistatics, Inc. (doing business as Ixana). This affiliation is disclosed as a potential competing interest. The catalogue includes publications coauthored by Shreyas Sen.

Inclusion requests and corrections

We may have missed relevant papers. If you would like a relevant paper published in a Tier-1 journal or conference to be considered for inclusion, please email shreyas@purdue.edu with its title, full citation, and DOI or source link. Requests will be reviewed for inclusion.

If you find a mistake, please email shreyas@purdue.edu with the paper identifier or DOI and a description of the error. Reported issues will be reviewed, and confirmed corrections will be incorporated into subsequent updates.

Acknowledgements

The author thanks Gourab Barik, Baibhab Chatterjee, Shovan Maity, Mayukh Nath, Debayan Das, Donghyun Seo, David Yang, and Arunashish Datta for their contributions to V1 and V2 of the survey.

5 Integrated circuits (103 records)

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